# Solar Energy

From lux on a sensor to watts on a roof — and from one measurement to three numbers calculated out of it

**Level:** Explorer and Scientist — ages 12–19 (Romanian school years VI–XII) • **Time in class:** 20 min, 50 min, or a 2–4 week project • **Sensors:** The light sensor (illuminance) and BME280 (air temperature)




*The activity screen. The values in the picture are simulated; your own will differ.*

## Before you start

**Who it is for**

Explorer and Scientist — ages 12–19 (Romanian school years VI–XII). The same app serves the whole range, but the task differs. At **Explorer** pupils read the illuminance, compare places and discover that tilting reduces the light collected. At **Scientist** they verify the cosine law numerically, criticise the conversion factor the app uses, and size a real installation. The most direct syllabus entries are in **year VII** (mathematics: the cosine of an acute angle), **year VIII** (energy, power, efficiency), **year IX** (geography: solar radiation) and **year XII** (semiconductors and the $p$–$n$ junction).

**Assumed knowledge**

Reading a chart and working out a ratio. For the cosine law: **the cosine of an acute angle**, so from year VII; in year VI the same investigation is done qualitatively, with the ratio of two readings. For the project: the arithmetic mean, an $XY$ chart in a spreadsheet and, at Scientist level, the area under a curve by the trapezium rule.

**Time in class**

Short: 20 min (demonstration). Full: one 50-minute lesson (one investigation carried through). Project: 10–20 school days at 10 min a day, plus two 50-minute analysis lessons.

| | |
|---|---|
| **Teacher preparation** | 20 min — check the box's firmware (with firmware older than 2.2.5 the sensor stops at about 3800 lx, so it cannot see direct sun); choose the fixed station and **check it at three times of day**, because a spot that is unshaded at 11 may be behind a tree at 14; record the orientation and the tilt; set up the shared folder; look up the school's latitude, which you need for the optimum-tilt discussion. |
| **Sensor settling time** | **20 seconds** for light at each position — the light sensor responds practically instantly, and the wait is only so that the hand which moved the box is out of the field. **But temperature needs 60–90 seconds**, and the **Panel temp.** and **Efficiency** cards depend on it. Anyone who moves the box into the sun and reads the efficiency immediately is reading the temperature of the shade. |
| **Sensors and boxes** | The light sensor for illuminance and BME280 for air temperature. **Both are needed**: without temperature the **Panel temp.** card stays empty, **Efficiency** stays at the reference value and the thermal penalty — the most interesting part of the activity — can no longer be seen. One box is enough for the demonstration and for the project; two make the tilt investigation much better, because one can stay horizontal as a control. |

**Three routes**

| | |
|---|---|
| **Short** | 20 min: projected demonstration — the box carried from the darkest corner to the window and then outdoors, with four orders of magnitude crossing the card, plus tilting and a hand shadowing the sensor. |
| **Full** | The 50-minute lesson: one investigation carried out properly — the tilt (investigation 2) or the lamp distance (investigation 3), with a prediction, a graph and a conclusion stated as a rule. |
| **Project** | 10–20 school days at 10 min a day, at a fixed station, with two analysis lessons at the end and the sizing of a real installation for the school roof. |

**The best time to start this project is September or March**

The daily peak illuminance changes fastest around the equinoxes. A four-week run starting in late September or late March shows a clear and unmistakable trend in peak light; the same run in June or December shows almost nothing. If you only have two weeks, choose them then.

## 1. What the app shows and what it does not

Unlike the guided activities, this is not a step-by-step wizard. It is an **open instrument**: there is no green button to advance, no **Back**, no **Finish** and no observation journal. **You set the pace of the lesson**, and the investigations in section 6 are ready to run exactly as written.

There are seven cards on screen, separated by the word **Model:**, and the difference between the two groups is the most important thing this activity has to teach. Before **Model:** stand the two *measured* quantities:

- **Illuminance**, in `lx` — straight from the light sensor.
- **Air temp.**, in °C — the temperature of the *air* around the box, not of a panel.

After **Model:** come five quantities *calculated* from those two:

- **Irradiance**, in W/m$^2$ — the illuminance divided by 120 ("for sunlight only", says the card itself).
- **Panel temp.**, in °C — the air temperature plus the typical warming of a panel in the sun (the NOCT rule).

- **Efficiency**, in per cent — 18% corrected for the panel temperature.
- **Power, 1 m²**, in W — what one square metre of panel would give now.
- **Energy, 1 m²**, in Wh — the power added up over time, since the window was opened.

Below the cards, a line of text, **Light: ...**, describes the light in words, by fixed illuminance thresholds.

The word **Model:** is not a formality: it means that everything after it is *worked out from something else*, with formulae. The core skill of this activity is the **critical use of a derived quantity**. The sensor measures lux. Everything else is calculated from lux by a model with assumptions hidden inside it. Pupils who can state those assumptions have understood the physics; pupils who read the W/m$^2$ card as a measurement have not.

**Technical note — what is on screen, what is computed, and what misleads**

**The sensors.** The light sensor for illuminance (the `LUX` column in the exported files) and BME280 for air temperature (the `T` column). Nothing extra is connected.
**How it starts.** From the **Activities** page, with the **Start** button on the **Solar Energy** card. The app works with a single box, whose name appears beside the title; if several suitable boxes are connected, the **Change box** list lets you pick another, and with no box the window shows **No box connected**. The cards refresh every second.
**The light sensor's range depends on the firmware.** With firmware 2.2.5 or newer, the box chooses its own sensitivity and measures up to about 120 000 lx — the whole range of this activity, from the corner of the room to summer sun. With older firmware the sensor **stops at about** 3800 **lx**: at the window and outdoors the number sticks, and the day's curve comes out cut off as a flat plateau. If you see that, reflash the box (the flashing guide in the `firmware` folder); the sensor is not faulty. A box still on the old USB-only firmware (1.x) gets a message from the app as well: flash the WiFi firmware, which works over USB too. On a box with old firmware the app shows a yellow warning by itself when the reading stops at about 3800 lx. Above about 10 000 lx the absolute values are approximate (roughly ±15–20%): the relative comparisons (tilt, sun against shade, day against day) count, not the exact lux.
**The formulae the app uses.** The irradiance is

$$E = \frac{L}{120},$$

with $L$ the illuminance in lux. The panel temperature is estimated with the NOCT rule (a typical panel reaches 45 °C in 20 °C air at 800 W/m$^2$):

$$T_{\text{panel}} \approx T_{\text{air}} + \frac{45 - 20}{800}\, E = T_{\text{air}} + 0.031\, E,$$

that is, some 25 °C above the air in full sun and almost nothing indoors. The efficiency is

$$\eta = 18\% \times \left[1 - 0.004\,(T_{\text{panel}} - 25)\right],$$

applied **on both sides of 25 °C**: a cold panel has a slightly *higher* efficiency than 18%, as in reality. Finally, $P = \eta \cdot 1\ \text{m}^2 \cdot E$, and the energy card adds up $P \cdot t$ at every new reading; gaps longer than 30 seconds are not added, and the total starts again from zero when the window is reopened or the box is changed. All the formulae also appear in the app's **Theory** tab. **If temperature is missing**, **Panel temp.** stays empty and the efficiency stays fixed at 18% — a sign that the BME280 is not reporting.
**The three tabs. Live Charts** holds two stacked charts: **Illuminance** and **Air temperature (the box, not a panel)** — the title itself says the box has no panel; a real panel in

the sun would be 20–30 °C warmer, as the **Panel temp.** card shows. **Lux → irradiance** shows the conversion's points, and **Theory** explains the solar cell (the $p$–$n$ junction, the photovoltaic effect), the model with its formulae and a few experiments.

**The conversion tab cannot show anything but a straight line** — and this remains the best question in the whole activity. Illuminance is on the horizontal axis and irradiance on the vertical, but the irradiance *is* the illuminance divided by 120. The points therefore fall exactly on a line through the origin whatever the data, even if the sensor is kept in a pocket.

The banner above the tab states plainly that this is a conversion rather than a measurement, and that the straight line is arithmetic rather than evidence. **The question to put to a class**: *what have you learned about the world from a chart in which a quantity is plotted against itself?* A pupil who answers it well has done the best possible piece of work on this activity.

**The Light line.** Below the cards the app writes **Light:** followed by a description. The full thresholds are: below 1 lx *Total darkness*; below 50 *Dim light (night / low interior)*; below 500 *Normal indoor lighting*; below 2000 *Overcast day / near window*; below 20 000 *Indirect sunlight*; below 80 000 *Direct sunlight (partial)*; above that, *Full direct sunlight*. The last two states appear only with the new firmware. Print this list and tape it beside the laptop.

**The chart buttons.** They sit in the right-hand corner of the tab row and show only on the **Live Charts** tab: **Smooth: on** / **Smooth: off**, **Pause** / **Resume**, and **Clear**. The pause button freezes *only* the drawing, not the recording: the data keep accumulating and the export will contain them. Smoothing, too, changes only the drawing; the exported file holds the raw values.

**The chart only shows the last two minutes.** For the whole-day investigation this means something very concrete: **the day's curve is not on the screen, it is in the exported file**. Do not look for the bell curve on the chart at three in the afternoon; it will not be there.

**The export, and its real limit.** The **Export data** button sits at the top right, beside **Close**. It opens a save dialogue with a suggested name that already contains the box and the date — the name given to the box in **Settings**, for example `Grupa_1_2026-09-14.xlsx`, or, if it has none, its code, for example `BOX_3859_2026-09-14.xlsx` — and writes an `.xlsx` (the default) or `.csv` file according to the extension chosen. On success a message gives the file path; if nothing has been recorded yet, the message says there are no data. The file holds the raw readings, with an `elapsed_s` column; the calculated quantities (irradiance, power) are *not* in it, so they are rebuilt in the spreadsheet with the same formulae. **The export contains at most** 72 000 **rows**, that is about 20 hours at one reading per second. A school day fits easily; **an unbroken run across a whole weekend does not**, and the earliest hours are lost silently. For runs longer than a day, export at the end of each day.

**The automatic log.** Independently of the button, the app writes a raw CSV log of the stream to the `logs` folder as the data arrive. That file does *not* carry the 72 000-row limit and can rescue a forgotten export. The app **no longer deletes logs by itself** (in **Settings**, **Keep logs (0 = always)** is 0 by default), but a `logs` folder is no substitute for the dated file you place in the project folder yourself.

## 2. Where this fits in the curriculum

- **Physics** — in **year VI**, measuring physical quantities and introductory optics (sources of light, straight-line propagation, shadow) support the demonstration route. In **year VIII** the heart of the activity settles: **energy, power and efficiency**, and $P = \eta A E$ is exactly the calculation pattern the syllabus asks for, with one quantity measured by the pupils instead of given in the question. In **year XII**, the photon, semiconductors and the $p$–$n$ junction explain *why* a silicon cell produces current when it is lit: light creates electron–hole pairs and the junction's field

separates them (the photovoltaic effect, not the external photoelectric effect, in which electrons are knocked out of a metal).

- **Mathematics — the cosine of an acute angle**, from **year VII**, is what makes the tilt investigation a verification rather than an impression. Direct and inverse proportionality also appear, along with the $XY$ chart and its line of best fit and, in **years XI–XII**, the area under a curve as accumulated energy — a rare and very concrete way into the idea of an integral.
- **Geography** — in **year IX**, general physical geography: solar radiation, the radiation balance, the angle of incidence and zonality. Latitude, season and the sun's elevation stop being definitions and become numbers the class has measured. In **years XI–XII**, renewable resources and the siting of a power station.
- **Technology and practical applications** — sizing a photovoltaic installation for a given demand, orientation and tilt as design decisions, the relationship between cost and output.
- **Computing and ICT** — processing the pupils' own dataset in a spreadsheet: filtering, mean, chart, trapezium rule. The data are theirs, which changes their attitude to the exercise entirely.

The core skill is the **critical use of a derived quantity**, and it is tested by a single question at the end of the lesson: *which of the seven cards on the screen are measured and which are worked out?*

## 3. Time plan

| Route | Time | What happens |
|---|---|---|
| **Demonstration** | **20 min** | The screen projected. The box is carried from the darkest corner of the room to the window and then outdoors, and the illuminance card crosses four orders of magnitude in front of the class. Then it is tilted away from the window and the cosine law appears; then a hand shadows the sensor. Export at the end. Enough to establish that light varies far more than the eye believes. |
| **Full** | **50 min** | One investigation carried out properly: the tilt (investigation 2) or the lamp distance (investigation 3). Written prediction, measurement, graph, and a conclusion stated as a rule. Do not attempt two in one lesson. |
| **Project** | **10–20 days**, 10 min a day | A fixed sensor on a fixed surface, read and exported at the same three times every school day, tracking the daily and seasonal march of the solar resource. It ends with the sizing of an installation for a real school roof. Two 50-minute analysis lessons. It runs beautifully across a change of season. |

## 4. Materials

| Item | Per group | Notes |
|---|---|---|
| MagicBox with the light sensor and BME280 | 1 | Both sensors are needed: without temperature **Panel temp.** stays empty and the efficiency stays fixed. |
| Laptop with the app | 1 | It travels with the group outdoors, or stays at the window for the fixed station. |
| Protractor and a stiff board | 1 | For the tilt investigation: a printed protractor taped to a clipboard lets an angle be set and held reproducibly. |
| Tape measure or metre rule | 1 | For the lamp investigation and for recording the sensor's height and position. |
| Desk lamp with a small bulb | 1 | For the inverse-square investigation. An LED torch works; a large diffuse panel does *not*. |
| Sticky labels | 3–6 | For the station's exact position and for the marked distances on the bench. |
| A safe fixed spot with light access | 1 / class | An unshaded south-facing sill or an outdoor bench in sight of a window. It must be **exactly the same spot every day**. |
| Extension lead or power bank | 1 | For long runs and for the outdoor station. |
| Small photovoltaic panel and multimeter | 1 (optional) | The real payoff: comparing a panel's measured output with the power the app predicts. |
| Shared folder or USB stick | 1 / class | All the dated files in one place. |

**Preparation.** Choose the fixed station position before day one and check it at three different times of day, looking for shadows — a spot that is unshaded at 11 may be behind a tree at 14, and the result is a mysterious daily dip in the data. Record the orientation (which compass direction the sensor faces) and the tilt, and write both on the station log. Confirm that the sensor lies flat and face up unless the investigation says otherwise. Look up your school's latitude; you need it for the optimum-tilt discussion.

## 5. Scientific background for the teacher

**Lux is not W/m$^2$, and that is precisely the point**

The light sensor measures **illuminance**, in lux. Illuminance is a *photometric* quantity: radiant power per unit area, *weighted by the sensitivity of the human eye*, which peaks in the green at 555 nm and falls to zero in the infrared and the ultraviolet.
**Irradiance**, in W/m$^2$, is a *radiometric* quantity: total radiant power per unit area at all wavelengths. A photovoltaic cell responds to irradiance across a broad band including the near infrared — exactly the part the eye and the sensor cannot see at all.
The ratio between them is the **luminous efficacy** of the source, in lumens per watt, and it depends entirely on the spectrum:

$$E\ [\mathrm{W/m^2}] \approx \frac{L\ [\mathrm{lx}]}{K},$$

where $K$ — lumens per watt *of radiation*, not per electrical watt — is about 100–120 lm/W

for daylight (a little less for the direct sun alone), only 15–20 lm/W for an incandescent lamp, because most of what it radiates is infrared, and around 300 lm/W for a white LED. **The app uses a fixed** $K = 120$, which is a reasonable approximation *for daylight and for nothing else.*
The consequence to teach: under the sun, the app's W/m$^2$ figure is an honest estimate. Under a white LED desk lamp it overestimates the irradiance by about two and a half times, because the LED puts nearly all its power where the eye is sensitive and almost none in the infrared. Under a filament bulb it underestimates it six- to eightfold, because the sensor does not see the infrared — which a silicon cell partly uses. **A conversion factor is not a constant of nature; it is a property of the light source.**

## 5.1 Reference numbers worth knowing

Noon sun at mid-latitudes gives around 1000 W/m$^2$ — the standard condition against which panels are rated — or about 100 000–120 000 lux. Daylight under overcast sky is 5000–20 000 lx, a well-lit classroom 300–500 lx, a corridor at night perhaps 50 lx. The eye compresses this enormous range logarithmically, which is exactly why pupils consistently underestimate how much darker the back of the room is than the window row.

## 5.2 The cosine law

If a beam of irradiance $E_0$ arrives at an angle $\theta$ to the normal of a surface, the power intercepted per unit area of that surface is

$$E = E_0 \cos\theta.$$

At $\theta = 0$ the surface faces the source squarely and collects everything; at 60° it collects half; at 90°, nothing. This single equation explains why panels are tilted, why winter yields are poor, why the poles are cold, and why morning and evening sun is weak even under a clear sky. It is the most productive formula in this guide, and pupils can verify it in ten minutes with a protractor.

For a fixed panel, the rule of thumb is a tilt roughly equal to the site latitude for the best annual yield — steeper for a winter bias, shallower for a summer one — facing the equator.

## 5.3 Daily and seasonal variation

Two effects combine. The sun's elevation changes through the day and through the year, so $\cos\theta$ on a horizontal surface changes; and when the sun is low its light traverses a longer atmospheric path — a greater *air mass* — so more is scattered and absorbed. Together these give the bell-shaped daily curve, narrower and lower in winter, broader and higher in summer. In Bucharest the total daily solar energy on a horizontal surface is on average **about** 4.5–5 **times** greater in June than in December (roughly 6 against 1.3 kWh/m$^2$ a day), and pupils who have measured even a fortnight of autumn find that entirely credible.

## 5.4 From irradiance to panel output, and the temperature penalty

The power delivered by a panel of area $A$ and efficiency $\eta$ is

$$P = \eta\, A\, E,$$

and the energy over a period is $W = P\,t$, conveniently in watt-hours.

Efficiency falls as the cell heats, at roughly 0.4% of output per degree above the reference temperature, and rises a little below it. The app models this with $\eta_0 = 18\%$, on both sides of 25 °C, applied to the estimated panel temperature. A panel at 60 °C — which a dark roof-mounted panel easily reaches in summer — has lost about 14% of its output. This produces the genuinely

counter-intuitive result that a cold, bright, clear winter day can be a better day, per watt-peak installed, than a sweltering August afternoon.

**To remember and to tell the class:** the box measures *air* temperature, not cell temperature, and a real cell in the sun runs 20–30 °C above the air around it. That is why the app does not use the air temperature directly, but **estimates** the panel's with the NOCT rule: +0.031 °C for every W/m$^2$, so about +25 °C in full sun. **That, too, is a model**: wind, the way the panel is mounted and its colour change the real figure a great deal, and saying so is a very good piece of critical thinking.

# 6. Ready-made investigations

## 6.1 Investigation 1: how much light is really in this room?

**Research question.** How far does illuminance vary inside a single classroom, and does it match what our eyes tell us?

**Method.** Before any measuring, every pupil writes a prediction: how many times brighter is the window sill than the darkest corner? Twice? Ten times? Then measure, with the sensor flat and face up at desk height, at eight positions across the room, holding each for 20 seconds. Repeat with the lights off and again with the blinds closed.

**What to measure.** Illuminance in lux at each position under the three lighting conditions; the ratio of the largest value to the smallest.

**Expected finding.** Ratios of 20:1 to 100:1 within a single room on a bright day, against typical pupil predictions of 2:1 to 5:1. Artificial lighting alone is far more uniform but much dimmer near the window. The lesson is about human perception as much as about light: the eye's logarithmic response hides an enormous range.

## 6.2 Investigation 2: the tilt angle — verifying the cosine law

**Research question.** How does the angle of a panel change the energy it collects?

**Method.** Mount the box on a stiff board with a protractor taped to the edge. Outdoors on a clear day, or indoors with a single lamp at a fixed distance, point the board straight at the source and record the illuminance. Then tilt to 30°, 60° and 75° — the four angles printed on the sheet — holding each for 20 seconds. Record the value at each angle and **return to** 0° **at the end**, to check that the source has not changed in the meantime. Four angles are enough to see the cosine, and they fit into one lesson; a quick class can add 15° and 45° on the back of the sheet.

**What to measure.** Illuminance at each angle $\theta$; the ratio $L(\theta)/L(0)$; and, plotted against $\cos\theta$, a straight line through the origin.

**Expected finding.** $L(\theta)/L(0)$ follows $\cos\theta$ closely outdoors under a clear sky — at 60° it should be about half. Under overcast sky the fall is much gentler, because most of the light is diffuse and arrives from the whole sky rather than from one direction. **The contrast between a clear-sky run and an overcast one is the most sophisticated result in this guide**, and both are worth doing. In year VI, where the cosine is not yet in the syllabus, the same investigation stops at the ratio and its wording: "the more we tilt, the less we get — and at 60° only about half".

## 6.3 Investigation 3: the inverse-square law, with a lamp

**Research question.** How does illuminance depend on distance from a small source?

**Method.** Indoors, in as dark a room as possible, place a small lamp at one end of the bench and mark distances of 20, 30, 40, 60, 80 and 120 cm. Place the box flat and facing the lamp at

each mark for 20 seconds. **Measure the background illuminance with the lamp off and subtract it** from every value.

**What to measure.** Illuminance at each distance $d$; then plot $L$ against $1/d^2$.

**Expected finding.** A straight line through the origin, confirming $L \propto 1/d^2$, provided the lamp is small compared with the distances used and the background has been subtracted. Deviation at short distances is a real and explainable effect — the source is not a point — and should be discussed rather than treated as error. Ask why the same law does *not* apply to the sun: at 150 million kilometres, a metre closer changes nothing.

### 6.4 Investigation 4: the daily solar curve

**Research question.** What shape does the solar resource have through a day, and when is the best time to generate?

**Method.** Leave the box at the fixed station, outdoors or at the window, horizontal and unshaded, on mains power or a power bank, for a whole school day. Export at the end. Repeat on a clear day and on an overcast one.

**What to measure.** Illuminance every few seconds through the day; the peak value and the time it occurred; the derived irradiance; the air temperature alongside. If the app's window stays open all day, the **Energy, 1 m²** card shows in the evening the day's energy according to the model, for one square metre of panel — a good number to check with the trapezium rule.

**Expected finding.** A smooth bell curve on a clear day, peaking near solar noon — often *not* at 12 on the clock, because of time zones and summer time, which is a satisfying discovery in itself. On a cloudy day, a ragged trace with brief spikes that can *exceed* the clear-sky value, when a bright cloud edge reflects extra light onto the sensor. The area under the curve is the day's total energy, and comparing areas is far more meaningful than comparing peaks.

**One practical warning:** the chart on screen holds only the last two minutes. **The day's curve is visible only in the exported file**, opened in a spreadsheet — not on the screen at the end of the day.

### 6.5 Investigation 5: sizing a real installation for the school

**Research question.** Could the school roof power the ICT room?

**Method.** Use the multi-day dataset to obtain a typical daily energy figure: either by the trapezium rule applied to the exported data, or, more simply, the mean irradiance multiplied by the number of daylight hours logged. Measure or estimate an available roof area. Apply $P = \eta A E$ with a realistic $\eta$ of 18%, then subtract a temperature penalty using the measured air temperature plus a stated allowance for cell heating.

**What to measure.** Mean daily irradiance from the pupils' own data; the roof area; the ICT room's demand, from the ratings on the equipment or from the site manager.

**Expected finding.** A number of the right order of magnitude, and a very concrete sense of scale: at 18% efficiency a square metre of panel at 1000 W/m$^2$ gives about 180 W, but the daily average over a full year at our latitude is a small fraction of that. Pupils almost always overestimate the yield before doing the sum and are surprised afterwards. If a real panel and multimeter are available, compare the predicted output with the measured one for the same square metre and account for the difference.

## 7. Running a multi-day project

## 7.1 A four-week schedule

| When | Who | What happens |
|---|---|---|
| Day 0 (prep) | Teacher | Fix and label the station, check it for shadows at three times of day, record its orientation and tilt, set up the shared folder. |
| Week 1, day 1 | Whole class, 50 min | Lux versus irradiance, the cosine law, the station protocol. Run the tilt investigation. Practise exporting and naming. |
| Weeks 1–4, daily | Duty pair, 10 min | Readings at 9, 12 and 15 at the fixed station, entered on the wall chart, plus one export a day. The sky condition written in words each time. |
| Week 2, one day | Duty pair | A continuous whole-day run, the box left recording from before school until after, on a power supply. |
| Week 3, one day | Duty pair | A second whole-day run, chosen deliberately for contrasting weather. |
| Week 4 | Whole class, 50 min | Combine the daily files. Plot peak illuminance against date. Compare the two whole-day curves and their areas. Complete the sizing calculation. |

## 7.2 Naming and storing the exported files

The name the app suggests already contains the box and the date — for example `BOX_3859_2026-09-14.xlsx`. With three readings a day, extend it:

`BOX_3859_2026-09-14_0900.xlsx`

Keep the date in `YYYY-MM-DD` form, so that sorting by name sorts by time, and keep all the files in one project folder. Maintain an index sheet with one row per export: date, time, sky condition in a single word, sensor orientation, and anything unusual. **That single word for the sky** — clear, hazy, broken, overcast, rain — is what makes the whole dataset interpretable later.

## 7.3 Keeping days comparable

- **The station is never moved.** Orientation, tilt, height and surroundings must be identical every day. If it is moved, the series starts again.
- **The same clock times — and beware of summer time.** A one-hour change shifts every reading relative to solar noon. Record the change in the index sheet and expect a step in the data.
- **Watch for new shadows.** A parked van, a growing tree, a banner on a fence, scaffolding — any of these produces a sudden unexplained drop. Look at the station with your own eyes, every day.
- **Keep the sensor's window clean.** Dust, fingerprints, rain spots and condensation all reduce the reading. Wipe it gently before each session and note if it was dirty.
- **Record the sky in one word, from the same list.** Without the sky column, a fortnight of numbers cannot be interpreted at all.

## 7.4 Weekends, holidays and handovers

Weekend gaps matter more here than in other projects, because the seasonal trend the class is trying to see is slow and every missing day makes it noisier. Two approaches work. First, if the station is indoors at a window and can safely be left powered, let it record through the weekend

— but **export at the end of each day**, because a single export at the end holds only the last 20 hours or so, and the first two days would be lost silently. Second, accept the gaps and plot against *date* rather than against sample number, so that the gaps are visible on the axis and the trend line stays honest.

Over a holiday, stop cleanly, export, and treat the resumption as a new block; do not join the two with a straight line. For handover between class groups, keep a laminated station log with the box: the exact position, orientation, tilt, height, reading times, the naming rule and the sky-condition word list, plus one line a day signed by the duty pair.

**The investigation, in one table**

**The question.** How much of the light falling on a surface is lost when the surface does not face the source squarely?
**The hypothesis.** Written **before** any measuring: at 60° we will receive _____ of the light at 0°, plus a reason. It is not changed afterwards.
**What I change.** One thing only: the **angle** between the board and the direction of the source.
**What I measure.** Illuminance in lux, read from the **Illuminance** card after 20 seconds of stillness.
**What I keep the same.** The same box, the same source at the same distance, the same position on the bench or in the yard, the same person reading, no shadow of a head or a hand over the sensor, and **the same background light** — which is why this investigation is not done beside a window the sun keeps entering and leaving.
**How many repeats.** Two readings at each angle, and without fail a **return to** 0° **at the end**. If the final value differs from the first by more than a few per cent, the source changed during the measurement and the series is not sound — outdoors, a cloud is the cause almost every time.
**What makes a reading valid — and what makes it an artefact.** Valid is a reading taken with the board held firmly at the angle read off the protractor, with the sensor unshaded and the background unchanged. An artefact is anything else: a board held in the hand wobbles by several degrees, and at large angles that matters a great deal; a pupil leaning over casts a shadow; and a passing cloud changes $E_0$ between two angles. **If the graph of $L(\theta)/L(0)$ against $\cos\theta$ is not a straight line through the origin under a clear sky, look for shadows and clouds first, not for a faulty sensor.**

## 8. Analysis after the activity

## 8.1 What the data typically looks like

| Condition | Typical illuminance | Note |
|---|---|---|
| Full direct summer sun | 80 000–120 000 lx | About 700–1000 W/m$^2$, using $K = 120$. With old firmware anything above about 3800 lx shows as 3800. |
| Direct winter sun, midday | 20 000–50 000 lx | Lower sun and a longer atmospheric path. |
| Bright overcast day | 5000–20 000 lx | Diffuse light; the cosine effect is much weaker. |
| Heavy overcast, winter | 1000–4000 lx | Still far brighter than any indoor lighting. |
| Just inside a sunlit window | 2000–20 000 lx | The glass cuts a noticeable fraction. |
| Desk in the window row / in the far corner | 300–500 / 50–200 lx | Guidance for written work is 300–500 lx; the far corner is often 20 to 100 times darker. |
| Ratio at 60° tilt, clear sky | 0.45–0.55 of the value at 0° | The cosine law, verified. |
| Air at 35 °C, full sun ($\approx$ 800 W/m$^2$) | Panel temp. $\approx$ 60 °C | The efficiency drops to about 15.5%, a loss of some 14% of output. |

## 8.2 Comparing two days, step by step

Open two dated exports and align them on *clock time*. Plot both illuminance traces on the same chart. Then compare **three summary numbers**, not individual readings:

1. **The peak illuminance** — the day's largest value.
2. **The time of the peak** — and how far it is from 12 on the clock.
3. **The area under the curve** — the total light accumulated. It is easily estimated in a spreadsheet by the trapezium rule: for each neighbouring pair of readings, their mean multiplied by the time between them, then the sum. It is the number that actually matters to a solar installation.

The instructive comparison is a clear day against a cloudy one. The cloudy day often shows a *higher* instantaneous peak than the clear one, when a bright cloud edge adds reflected light to the direct beam — and yet its area is far smaller. **Peaks mislead; areas do not.** A pupil who understands that distinction has understood the difference between power and energy, which is the central idea of the whole topic.

## 8.3 Questions to ask afterwards

1. The app converts lux into W/m$^2$ by dividing by 120. Under what light source is that a good approximation, and under what source is it badly wrong? Explain using the eye's sensitivity curve.
2. The conversion tab shows points lying perfectly on a straight line, and says itself that they cannot do otherwise. What have you actually discovered about the world from that chart?
3. Your cloudy day had a higher peak than your clear day, but the clear day produced far more energy. Explain the difference between those two statements.
4. The efficiency card says a panel does worse when it is hot. Why, then, are there so many solar farms in deserts?

5. The tilt investigation matched $\cos\theta$ outdoors in sun but not under cloud. What does that tell you about where the light was coming from in each case?
6. Your peak value fell by 30% over three weeks in October. How much of that is the season and how much could be something else? Name two other possible causes and say how you would rule each one out.

## 8.4 Assessment ideas

- **Low stakes:** the pupil annotates one exported daily curve, marking the peak, estimating the time of solar noon and labelling one stretch where cloud passed.
- **Standard:** a report on the tilt or distance investigation, with a graph of illuminance against $\cos\theta$ or against $1/d^2$, a line of best fit, and a statement of how well the data support the law.
- **Extension:** the full sizing calculation from the class's own data, with the temperature correction applied, an explicit critique of the $K = 120$ assumption and a stated uncertainty on the final answer. The strongest pupils also explain *why* **Panel temp.** is an estimate and *why* the conversion tab could never have shown anything but a straight line.

### Common pitfalls and misconceptions

**Reading the W/m$^2$ card as a measurement.** It is the illuminance divided by 120. It is an honest estimate in daylight, about two and a half times too high under LED lighting and several times too low under a filament bulb. Say so on day one and keep saying it.
**Direct sun on the body of the box.** The sensor is meant to be in the sun — but the box warms quickly, and the temperature it then reports describes sun-heated plastic, not air. The **Panel temp.** and **Efficiency** cards, which start from that temperature, become meaningless. Shade the *body* while leaving the sensor's window exposed, or record temperature separately from a box kept in shade.
**Forgetting the sensor's orientation.** A light sensor is directional. A box lying flat, one propped at an angle and one standing on end give three very different values in the same place. Fix the orientation and record it.
**Shadows nobody noticed.** A hand, a head leaning over, a bag on the bench, a parked car. Half the anomalies in this dataset are shadows. Look before you record.
**Not subtracting the background in the lamp investigation.** Room lighting adds a constant offset that flattens the inverse-square relationship until it is unrecognisable. Measure with the lamp off and subtract.
**Comparing peaks instead of areas, with no note of the weather.** A cloudy day can out-peak a clear one, so compare the area under the curve — and a fortnight of numbers with no sky column cannot be analysed at all. One word per reading is enough.
**Expecting the sun to peak at 12.** The time zone, the position within it and summer time all shift solar noon, often by more than an hour. In Bucharest, in summer, solar noon falls at about 13:15 (summer time), a quarter of an hour earlier or later depending on the date.
**Looking for the day's curve on the screen.** The chart holds only the last two minutes. The day's curve exists only in the exported file.
**A perfectly flat plateau at about 3800 lx.** Not a strange sky but old firmware: the sensor cannot climb higher. Reflash the box; the outdoor data taken until then are cut off.
**One export after a run of several days.** The export takes at most 72 000 rows, about 20 hours. Anything older is missing from the file, with no warning at all. Export at the end of each day.

### Safety and classroom management

**Nobody looks directly at the sun**, and no optical instrument is pointed at it. This is the one genuinely serious hazard in the activity and it is stated explicitly before anybody goes outside. The sensor is pointed at the sun; the pupils are not.

**Heat and sunburn.** Outdoor work on bright days follows the school's sun policy: hats, shade breaks, water. Note also that dark surfaces and equipment left in full sun become hot enough to be unpleasant to handle.

**Equipment left unattended.** A station left recording outdoors or on a sill must be agreed with site staff, labelled with the class and teacher name, secured so that it cannot fall or be knocked, and never left in an escape route. Do not leave equipment outdoors overnight or when rain is forecast.

**Weather and cables.** Electronics and laptops are not used in the rain; bring the station indoors if the forecast changes. Extension leads must not cross walkways and must not be run outdoors through a door that has to stay closed — use a power bank for an outdoor station.

**Lamps.** The desk lamp used for the inverse-square investigation gets hot. Keep it clear of paper, let it cool before moving it, and check its lead.

**Roofs.** Nobody goes onto a roof. Roof area for the sizing calculation is estimated from a plan or from the ground, never by climbing.

### Adaptations for pupils with special educational needs

**A ranking task instead of a calculation.** The pupil reads the illuminance card and the **Light:** line at five given places and puts them in order from brightest to darkest, then says in one sentence where they would put a solar panel and why. The objective — light varies a great deal and can be compared — is met in full, with no formula at all.

**Axes drawn in advance.** For pupils whose whole lesson is consumed by constructing a graph, give a sheet with the axes already drawn, scaled and labelled, and three points to place. The idea being learned is the *shape* of the relationship, not the drawing of axes.

**A pre-filled column.** In the tilt investigation the $\cos\theta$ column is already printed for every angle. The pupil does the division and the comparison, which is the skill being assessed; looking up a cosine on a calculator is not.

**Three real roles for pupils with motor difficulties.** Not being able to hold the board does not mean not taking part. The roles rotate at every angle:

- **the angle keeper** — reads the protractor and gives permission to measure; they have the right to stop a reading taken with the board wobbling, so they hold the validity of the whole series in their hands;
- **the screen reader** — says the illuminance aloud, rounded, and announces when the 20 seconds have passed;
- **the scribe** — writes the values into the table and works out the ratio $L(\theta)/L(0)$ as they arrive, which makes the outliers visible immediately rather than at home.

**Sensory sensitivity and light.** A few pupils find bright light unpleasant, and a few are photosensitive. Do not send them to the outdoor measurements on very sunny days, and do not point the lamp in their direction; the scribe and screen reader roles are just as valuable and can be done from a desk in the shade.

**The long project as a strength.** The ten-minute daily routine, with the same steps in the same order, is for many pupils the most accessible part of the activity rather than the hardest. A duty pair with a laminated checklist is doing real, verifiable scientific work.

**If things do not go to plan**

**It is overcast all week and you never see the sun.** Do not postpone. The lamp investigation (the third) is done entirely indoors and gives the cleanest graph in the guide. And the tilt investigation under overcast sky is not a failure but *the other half* of the result: the gentle fall proves that the light is arriving from the whole sky. Do it and record it as such, and when a clear day comes, repeat it and put the two series side by side.

**You have no access outdoors or to a safe sill.** The station can stand indoors beside the glass, horizontal, in the same place. The values will be smaller, but the seasonal trend — which is the point of the project — survives intact.

**The efficiency hardly moves in class.** That is normal: indoors the irradiance is small, so the "estimated" panel is almost at air temperature, and the efficiency stays close to 18% — a little above, if the room is below 25 °C. The penalty shows outdoors, in the sun, where **Panel temp.** climbs some 25 °C above the air. If **Efficiency** stays fixed at 18% and **Panel temp.** is empty, then the BME280 is not reporting, and that is a problem with the box.

**You forgot to export and closed the app.** Look in the `logs` folder for that day's raw CSV. It carries no 72 000-row limit, so it may be more complete than the export. The app does not delete it by itself unless somebody has set a number of days under **Keep logs** in **Settings**; copy it into the project folder now anyway, not next month.

**The series has a missing day or an unexplained step.** Do not delete it and do not "correct" it. Note it in the index sheet and plot it as it is: hunting for the cause — a parked van, a lowered blind, the clocks changing — is a better lesson than a clean series.

**The box will not connect.** Do not stop the lesson. Use the "What the data typically looks like" table in this guide as a printed dataset: pupils write their predictions first, then receive the values row by row, work out the ratio at 60° and compare it with $\cos 60° = 0.5$. The discussion about the difference between a measured and a calculated quantity is unchanged — and it is the central lesson anyway. Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

## 9. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **National Renewable Energy Laboratory** — `https://www.nrel.gov` — search for solar resource data and reference spectra; the authoritative source for the 1000 W/m$^2$ standard test condition and for typical yields by location.
- **Copernicus** — `https://www.copernicus.eu` — European surface solar radiation data, so pupils can compare their fortnight of readings against a regional record.
- **Romanian National Meteorological Administration** — `https://www.meteoromania.ro` — search for hours of sunshine and cloud cover; useful for explaining a particular day in the class's series.
- **International Renewable Energy Agency** — `https://www.irena.org` — global solar capacity and cost data, for putting the class's sizing calculation into a real-world context.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — simulations on light, forms of energy and the photoelectric effect, which support the photovoltaic mechanism.
- **European Space Agency** — `https://www.esa.int` — solar power on spacecraft, where the cosine law and panel orientation are matters of survival rather than economics; excellent for engagement.

## 10. Assessment

**The exit ticket**, identical to the one on the pupil sheet: *which of the numbers on the screen were measured and which were worked out?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Explorer and Scientist level the three axes look like this:

- **Measurement practice** — the pupil fixes and records the sensor's orientation, respects the 20 seconds, subtracts the background where it is needed, and performs the control return to 0°.
- **Data interpretation** — the pupil plots the right quantity on the right axis ($\cos\theta$, not $\theta$; $1/d^2$, not $d$), compares areas rather than peaks, and reads a trend from a series with gaps in it.
- **Scientific reasoning** — the pupil distinguishes the measured quantity from the calculated one, states the $K = 120$ assumption and says under what source it fails, and explains why **Panel temp.** is still a model, not a measurement.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

## 11. Pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/solar_energy.pdf`. It has two pages — the station log, the prediction and the tilt investigation on the first; the two-day comparison and the conclusion on the second — so it is printed double-sided, one sheet per pupil or per duty pair. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Four things to know before printing it. **The first layer asks the pupil to tick which cards are measured and which are calculated**, and this is not a formality: it is the core skill of the activity, placed at the start, where it is also learned. **The $\cos\theta$ column in the tilt table is already printed** for every angle, so that the pupil does the ratio and the comparison — the thinking — without losing time on a calculator, and so that the investigation is accessible to those who do not yet have the cosine in their syllabus. **The last row is the control return to** 0°, without which the series cannot be declared sound. **The two-day comparison table asks for the area, not just the peak**, because that is precisely the difference between power and energy that the activity teaches.

**The sheet carries no daily-readings table**, although the project has one. That is a class document, not a pupil one: the daily readings live on the wall chart, in the project index sheet and in the exported files, and a few rows on every child's sheet would not have been a register but the beginning of one — taking exactly the space the two investigations the pupil actually reasons about need. The tables are not repeated in this guide precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.